ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

TECHNICAL SCIENCES

PAPER 1 (PHYSICS)

MARKS: 100 **TIME: 2 hours**

Topics: units and measurement, motion, energy and power, waves and sound, electric circuits, magnetism.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show formulae, substitutions and units in ALL calculations. Use g = 9,8 m·s⁻².

$v_f = v_i + a\Delta t$	$\Delta x = \frac{v_i + v_f}{2}\Delta t$	$W = F\Delta x$	$P = \frac{W}{\Delta t}$
$E_P = mgh$	$v = f\lambda$	$R = \frac{V}{I}$ $P = I^2R$	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$

QUESTION 1: MULTIPLE CHOICE

1.1 The SI unit of force is the …

A joule. B newton.
C watt. D pascal.

1.2 Which instrument measures the diameter of a thin wire most accurately?

A Metre rule B Micrometer screw gauge
C Measuring tape D Ammeter

1.3 The rate at which work is done is called …

A energy. B force.
C power. D momentum.

1.4 An ammeter must be connected in a circuit …

A in parallel. B in series.
C either way. D across the battery only.

1.5 Sound travels FASTEST through …

A air. B water.
C steel. D a vacuum.

(5 × 2)
[10]

QUESTION 2: UNITS AND MEASUREMENT

2.1 Write down the SI unit of: (a) mass; (b) time; (c) electric current. (3)

2.2 Convert: (a) 2,5 kN to N; (b) 450 mm to m; (c) 72 km·h⁻¹ to m·s⁻¹. (4)

2.3 Write 0,000 32 m in scientific notation. (2)

2.4 On a micrometer screw gauge, the main scale shows 5,5 mm and the thimble scale shows 28 divisions. Each thimble division is 0,01 mm. Calculate the reading. (3)

[12]

QUESTION 3: MOTION

3.1 A delivery truck drives 12 km north and then 5 km south. The trip takes 20 minutes.

3.1.1 Write down the total distance travelled. (1)

3.1.2 Write down the truck's displacement. (2)

3.1.3 Calculate the average speed in km·h⁻¹. (3)

3.1.4 Calculate the average velocity in km·h⁻¹. (3)

3.2 A car accelerates uniformly from 10 m·s⁻¹ to 30 m·s⁻¹ in 5 s.

3.2.1 Define *acceleration*. (2)

3.2.2 Calculate the acceleration of the car. (3)

3.2.3 Calculate the distance covered in the 5 s. (3)

3.3 Explain the difference between a scalar and a vector, with ONE example of each. (3)

[20]

QUESTION 4: ENERGY AND POWER

An electric hoist in a workshop lifts a 200 kg engine vertically through 3 m at a constant speed in 12 s.

4.1 Calculate the weight of the engine. (2)

4.2 Calculate the work done by the hoist on the engine. (3)

4.3 Define *power* and calculate the useful power of the hoist. (5)

4.4 Write down the gravitational potential energy gained by the engine. (2)

4.5 The hoist is only 70% efficient. Calculate the electrical power it uses. (3)

4.6 State the principle of conservation of energy, and name the main energy change in the hoist's motor. (3)

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QUESTION 5: WAVES AND SOUND

5.1 Explain the difference between a longitudinal and a transverse wave. Give ONE example of each. (4)

5.2 A sound wave has a frequency of 200 Hz and a wavelength of 1,7 m. Calculate its speed. (3)

5.3 Explain how ultrasound can be used to find cracks inside a metal part. (2)

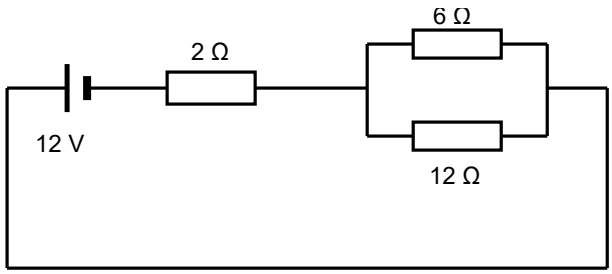
5.4 Explain why workers must wear hearing protection near loud machines. (3)

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QUESTION 6: ELECTRIC CIRCUITS

A 12 V battery (no internal resistance) is connected to a 2 Ω resistor in series with a parallel combination of a 6 Ω and a 12 Ω resistor.

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6.1	Define *resistance*.	(2)
6.2	Calculate the effective resistance of the parallel part.	(3)
6.3	Write down the total resistance of the circuit.	(1)
6.4	Calculate the current from the battery.	(3)
6.5	Calculate the potential difference across the 2 Ω resistor.	(2)
6.6	Calculate the current through the 12 Ω resistor.	(3)
6.7	Calculate the power dissipated in the 2 Ω resistor.	(3)
6.8	Explain how a fuse protects an electrical appliance.	(3)
		[20]

QUESTION 7: MAGNETISM

7.1	Draw the magnetic field pattern around a single bar magnet. Show the direction of the field lines.	(3)
7.2	Give TWO ways to make an electromagnet stronger.	(2)
7.3	Give ONE industrial use of an electromagnet.	(1)
7.4	Name TWO ferromagnetic materials.	(2)
		[8]

TOTAL: 100

END OF QUESTION PAPER